EV EBIT and GPOA standalone

Automated factor-model test

Practitioner factor-model runner

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Abstract

The run evaluates 2 custom factors (EV_EBIT, GPOA) on 4480 symbols in the full screened universe from gold version=20260822T141457Z. The strongest net factor is GPOA (annual return 0.51%, Sharpe 0.09, HAC p=0.746); 1 of 2 factor means have p<0.05 before any search adjustment. The report is generated from the exact configuration and measured outputs of this run; it is an exploratory model diagnostic, not evidence from an untouched confirmatory sample.

1 Research objective

This run asks whether the configured composite signals produce useful factor-mimicking returns, whether those returns are explained by the supplied benchmark or standard factor model, and whether each explicitly paired candidate improves its baseline. Signals are observed at month-end and evaluated on next-calendar-month simple returns.

Reading rule. Standalone performance asks whether the factor earns a premium; spanning alpha asks whether configured controls explain that premium; a custom-versus-custom paired test rebuilds both portfolios on identical stock-months before asking whether a proposed composite improves its named baseline. An external-return baseline can only be matched by month. Correlation and PCA describe overlap but do not replace the spanning test.

2 Configuration and sample

Factor	Composite signals	Signal transform	Composite transform	Winsor	Min inputs
EV EBIT	earnings_yield	robust	rank	5%-95%	1
GPOA	gpoa	robust	rank	5%-95%	1

Factor	Construction	Book	Beta neutral	Group relative	Bucket frac.	Cost bps
EV EBIT	signal weighted	long-short	yes	sector (signal)	0.33	10.0
GPOA	signal weighted	long-short	yes	sector (signal)	0.33	10.0

The screened sample contains 4,480 symbols and 274,708 stock-months from 2002-03-01 through 2026-08-01. Universe settings are: minimum as-traded price 5.0, minimum 21-day dollar volume 1000000.0, minimum market capitalization 250000000.0, top-cap fraction None, and financial exclusion True. External return controls are benchmark file none and standard-factor file standard_recreated_factors.csv. The runner reports the supplied files verbatim and does not infer their provenance.

3 Standalone factor performance

Return statistics use each factor's available months. Annual return and volatility use monthly arithmetic moments, Sharpe is their ratio, and drawdown is measured from the compounded net-return path. The cumulative-return figure uses the longest window shared by every custom factor; spanning and pair tests use matched months only.

Factor	Return months	Mean stocks	Min stocks	IC months
EV EBIT	235	891.7	20	235
GPOA	235	953.3	25	235

Factor	Ann. ret.	Ann. vol.	Sharpe	Max DD
EV EBIT	-4.67%	7.55%	-0.62	-64.20%
GPOA	0.51%	5.87%	0.09	-30.42%

Common-window performance uses 235 identical months from 2007-02-01 through 2026-08-01.

Factor	Common N	Ann. ret.	Ann. vol.	Sharpe	Max DD
EV EBIT	235	-4.67%	7.55%	-0.62	-64.20%
GPOA	235	0.51%	5.87%	0.09	-30.42%

Factor	HAC t	p	Mean IC
EV EBIT	-2.83	0.005	0.0038
GPOA	0.32	0.746	0.0113

Factor	Mean gross	Max abs net	Max abs beta	Mean turnover
EV EBIT	1.965	1.25e-15	1.76e-15	0.211
GPOA	1.975	9.27e-16	1.14e-15	0.131

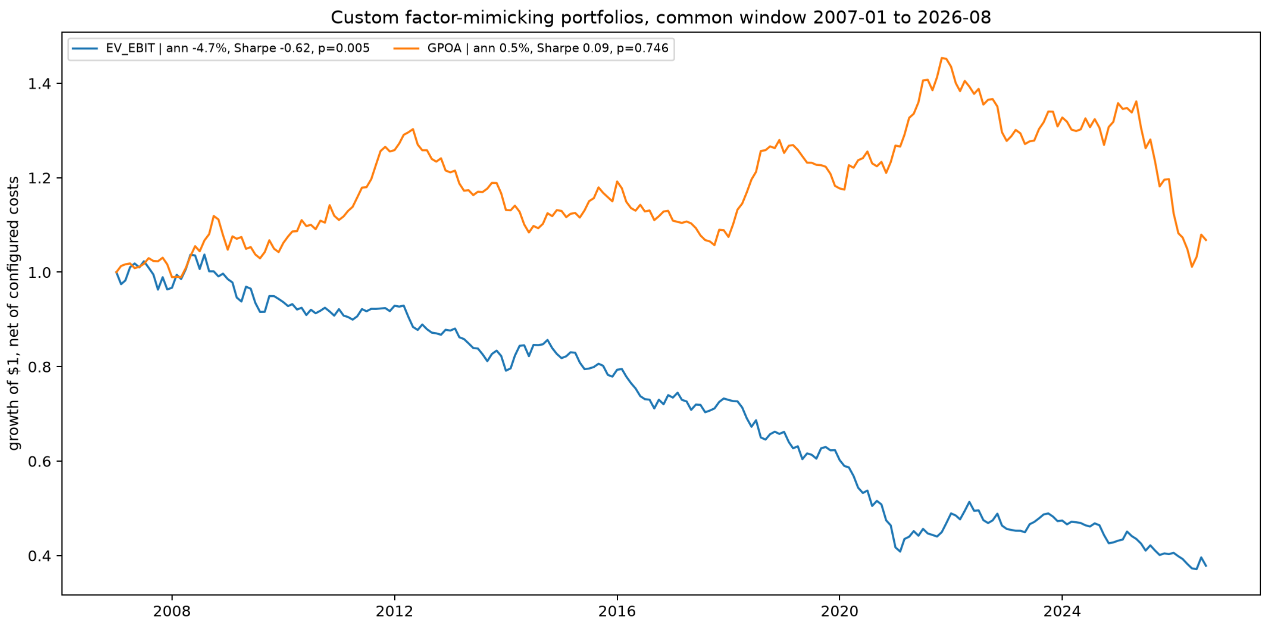


Figure 1: Net factor-mimicking-portfolio growth on the longest shared monthly window. Every series begins at one on the same date.

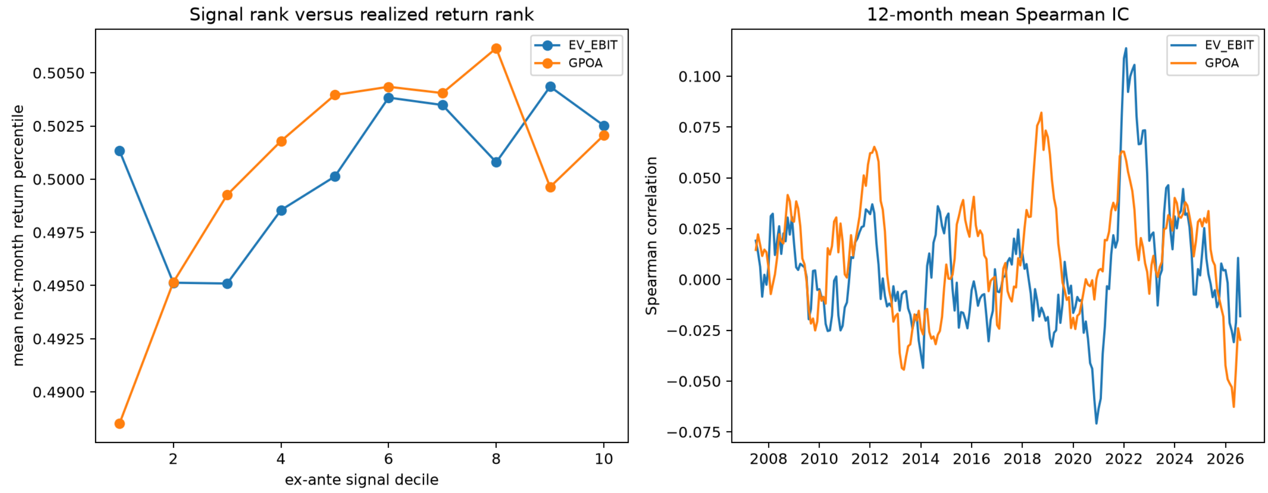


Figure 2: Ex-ante factor ranks versus next-month realized return ranks, and the time variation in monthly Spearman information coefficients.

4 Incremental return versus configured controls

Alpha is the annualized intercept from a monthly time-series regression on every supplied control; its HAC t and p test whether residual mean return remains. R2 measures the share of monthly return variation explained, while the joint F test asks whether the control slopes are jointly zero.

Factor	**N**	**Alpha**	**t**	**p**	**R2**	**Joint F**	**F p**
EV EBIT	230	-5.59%	-3.91	0.000	0.355	33.48	0.000
GPOA	230	-0.74%	-0.49	0.626	0.198	8.35	0.000

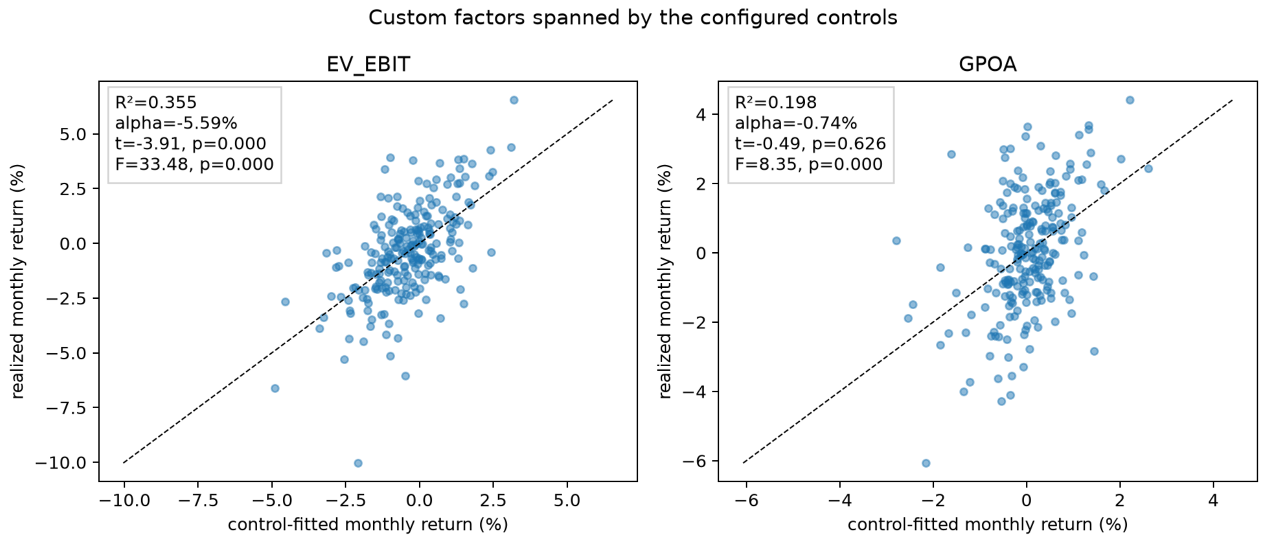


Figure 3: Realized custom-factor returns against returns fitted by the configured benchmark or standard-factor controls. Dashed lines are 45-degree identity references; no horizontal or vertical zero guides are drawn.

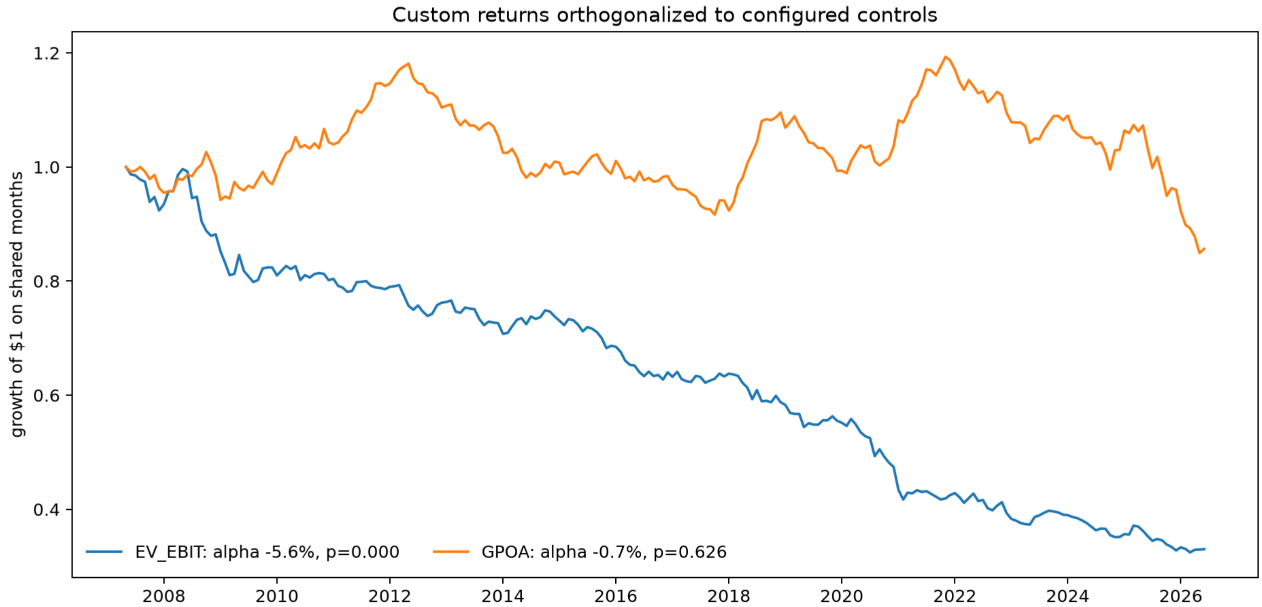
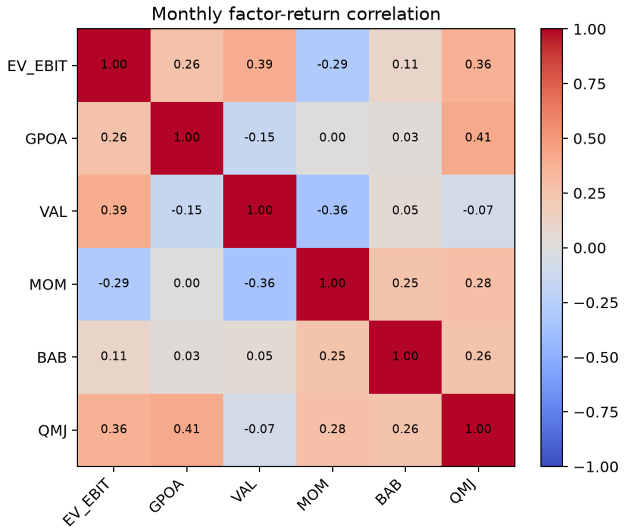


Figure 4: Cumulative alpha-preserving custom-factor returns after removing fitted exposures to the configured benchmark or standard-factor controls, aligned to one shared monthly window.

5 Correlation and principal components

The joint correlation and PCA include the standard factor returns when supplied. PC1 explains 30.4% of standardized common-window variation across 230 months. A large custom loading beside a standard factor with the same sign is descriptive overlap, not by itself proof of spanning; the regression above provides the alpha test.



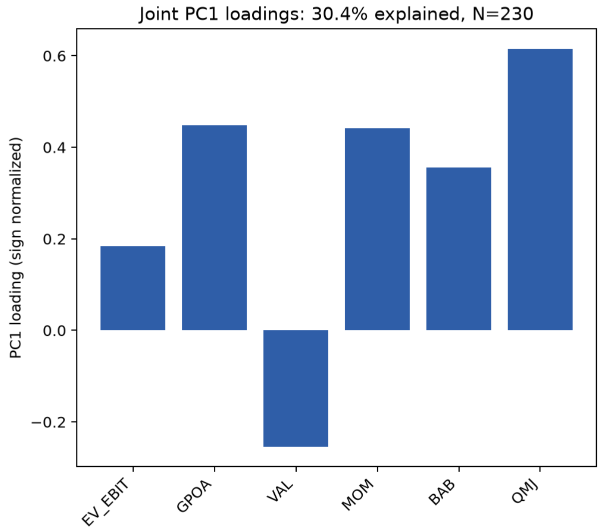


Figure 5: Monthly return correlations and sign-normalized first-principal-component loadings for custom and supplied standard factors.

6 Candidate-versus-baseline tests

Return change is candidate minus baseline on the displayed matched population; its HAC t and p test whether that mean improvement differs from zero. “Mean names” is the average common stock count before portfolio construction and is available for custom-custom pairs. Candidate alpha is estimated against the named baseline.

Comparison	Population	Months	Avg names	Return diff.	t	p	Return corr.
EV EBIT versus standard VAL	months; external return	235	n/a	-4.03%	-1.46	0.145	0.391
GPOA versus standard QMJ	months; external return	232	n/a	-7.58%	-2.41	0.017	0.409

Comparison	Cand. alpha	Alpha t	Alpha p	R2
EV EBIT versus standard VAL	-4.32%	-2.82	0.005	0.153
GPOA versus standard QMJ	-0.87%	-0.53	0.595	0.167

Comparison	Exposure corr.	IC change	IC p
EV EBIT versus standard VAL	n/a	n/a	n/a
GPOA versus standard QMJ	n/a	n/a	n/a

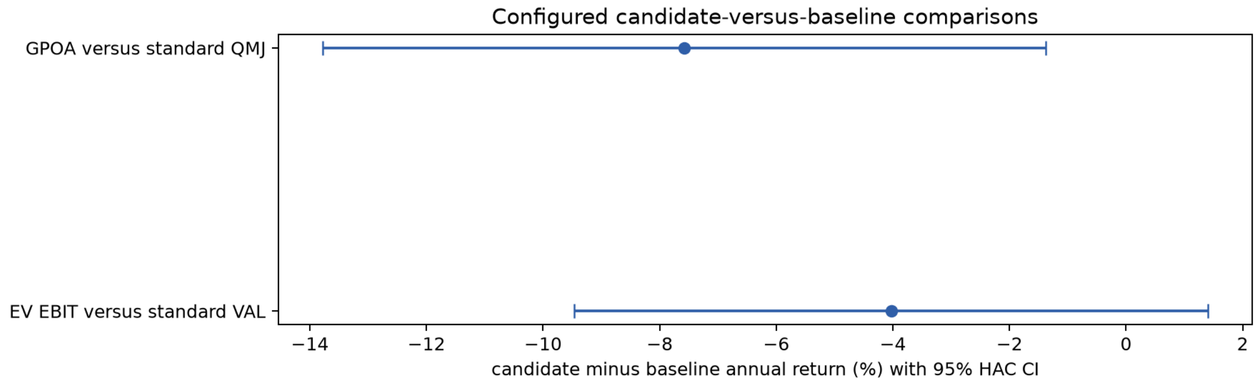


Figure 6: Annual candidate-minus-baseline return changes with Newey-West 95 percent confidence intervals on the configured matched population.

7 Conclusion

The run evaluates 2 custom factors (EV_EBIT, GPOA) on 4480 symbols in the full screened universe from gold version=20260822T141457Z. The strongest net factor is GPOA (annual return 0.51%, Sharpe 0.09, HAC p=0.746); 1 of 2 factor means have p<0.05 before any search adjustment. Benchmark alpha, joint PCA, and the configured pair tests should be read together: standalone performance answers whether a factor worked; spanning asks whether it is a new return direction; the paired comparison asks whether the proposed composite improved the specific existing factor it was designed to replace or extend.

8 Reproducibility

Run ID:

`20260824T071027Z_ev-ebit-and-gpoa-standalone_653d3463c1`

Configuration hash: `653d3463c1`. Gold version: `version=20260822T141457Z`. The run directory contains the normalized configuration, data manifest, exposures, factor returns, statistics, regression outputs, figures, LaTeX source, compiler log, and this PDF. Re-running never overwrites an existing directory.